

Research Profile

Researcher with 5+ years of experience in solving **nonlinear partial** and delay differential equations, conducting **stability** analysis, addressing **inverse problems**, and analyzing **stochastic** dynamics. My research develops computational and data-driven approaches to extract physical parameters and dynamical behavior from complex systems, with particular emphasis on **physics-informed neural networks** (PINNs) and numerical reconstruction.

My interdisciplinary background in Physics, nonlinear dynamics, computational modeling, and machine learning has enabled me to work across mathematical and engineering problems. My recent research applies PINNs to the data-driven reconstruction of dynamical states and estimation of unknown physical parameters in time-delayed problems, providing a direct connection between mathematical models, experimental data, and inverse parameter identification.

Education

- **Ph.D in Mechanical Engineering – University College Dublin, Ireland**

Jan 2023 – Nov 2026 | Expected Viva Date: Feb 2027

Thesis Title: Nonlinear Phenomena and Machine Learning Approaches for Nonsmooth and Time-delayed Machining Dynamics

Summary: Development of mathematical, numerical, experimental, and machine-learning-based approaches for understanding complex nonlinear and stochastic dynamical systems arising in machining. The research investigates regenerative and frictional effects, multistability, stochastic dynamics, and data-driven reconstruction and parameter estimation in machining using PINNs.

- **M.Sc in Physics – Bharathidasan University, India**

Jul 2018 – May 2020

Dissertation Title: Poisson Algebras for Classical Superintegrable Systems and Associative Algebras for Quantum Superintegrable Systems

Summary: This research provided a foundation in mathematical physics, analytical modelling, and the mathematical description of nonlinear and integrable systems.

- **B.Sc in Physics – Bharathiar University, India**

Jun 2015 – May 2018

Research Experience

- **Ph.D Researcher – University College Dublin, Ireland (Jan 2023 – Present)**

- Developed mathematical and numerical models of nonlinear, stochastic, and delay-dependent machining systems.
- Developed physics-informed machine-learning approaches for reconstructing dynamical states.
- Investigated inverse estimation of unknown physical parameters from dynamical data.
- Applied numerical continuation, bifurcation, stability, and time-integration methods.
- Conducted computational and experimental investigations of nonlinear mechanical systems.

- **Research Intern – AI Research Lab, Saama Technologies, India (Aug 2022 – Dec 2022)**

- Contributed to interdisciplinary AI research projects focused on improving clinical-trial processes through machine learning.
- Supported the development and evaluation of machine-learning approaches for applied problems such as the Multi-Component Manakov Model.
- Conducted data analysis and research to support the selection and evaluation of appropriate modeling approaches such as active learning sampling.
- Collaborated with researchers from different technical backgrounds to translate domain-specific problems into data-driven modeling tasks and communicate technical findings.

- **Project Fellow – Bharathidasan University, India (May 2021 – Apr 2022)**

- Worked on the application of machine learning and computational methods to nonlinear dynamical systems, with a particular focus on data-driven prediction and reconstruction of complex dynamical behavior.
- Developed and analysed the physics-informed neural network framework to approximate the data-driven solution of the nonlinear Schrödinger equation.

List of Publications

1. Jaganathan, M., Pakrashi, V., & Rounak, A., Physics-informed neural networks for parameter estimation in machining dynamics with regenerative delay and nonsmooth friction, manuscript in preparation (2026).
2. Jaganathan, M., Pakrashi, V., & Rounak, A., Interplay of nonsmoothness, time delay, and stochasticity in turning dynamics, *Communications in Nonlinear Science and Numerical Simulation*, 110149 (2026).
3. Jaganathan, M., Reilly, D., Nolan, K., & Rounak, A., Experimental insights into orthogonal cutting via high-speed camera tests, *IFAC-PapersOnLine*, 59(21), 107 – 112, (2025).
4. Jaganathan, M., Bakthavatchalam, T. A., Vadivel, M., Murugan, S., Balu, G., Sankarasubbu, M., Ramaswamy, R., Sethuraman, V., & Malomed, B. A., Data-driven multi-valley dark solitons of multi-component Manakov model using physics-informed neural networks, *Chaos, Solitons & Fractals*, 172, 113509, (2023).
5. Jaganathan, M., Manikandan, K., Sudharsan, J. B., & Senthilvelan, M., Data-driven soliton solution of the nonlinear Schrödinger equation with certain PT-symmetric potentials via deep learning, *Chaos*, 32(5), (2022).
6. Jaganathan, M., Sudharsan, S., Venkatesan, A., & Senthilvelan, M., Prediction of occurrence of extreme events using machine learning, *The European Physical Journal Plus*, 137(1), 16 (2021).
7. Jaganathan, M., Sudharsan, S., & Senthilvelan, M., Model-free prediction of emergence of extreme events in a parametrically driven nonlinear dynamical system by deep learning, *The European Physical Journal B*, 94(8), 156 (2021).

Conference Proceedings

1. Bakthavatchalam, T. A., Murugan, S., Vadivel, M., Jaganathan, M., Balu, G., & Sankarasubbu, M., Active learning for probabilistic machine learning-based modeling of dynamical systems, in *International Conference on Nonlinear Dynamics and Applications*, 399 – 408, Springer Nature Switzerland, (2023).
2. Jaganathan, M., & Senthilvelan, M., Prediction of chaotic attractors in quasiperiodically forced logistic map using deep learning, in *Nonlinear Dynamics and Applications: Proceedings of the ICNDA 2022*, 649 – 657, Springer International Publishing, (2022).

Skills & Competencies

- **Inverse modeling and parameter estimation:** Developing physics-informed neural-network methods to estimate unknown parameters in machining models from observed dynamical data. My current doctoral work treats parameter recovery as an inverse problem constrained by governing equations, regenerative time delay, and nonsmooth frictional contact.
- **Nonlinear dynamical systems:** Experience analyzing nonlinear, stochastic, delayed, and nonsmooth systems, including stability, multistability, bifurcations, chatter vibrations, chaotic attractors, and extreme events.
- **Differential equations and numerical modeling:** Experience working with ordinary, delay, and partial differential equation models. My research on nonlinear Schrödinger and Manakov systems has involved data-driven and physics-informed solution reconstruction, while my doctoral work uses time integration, numerical continuation, and bifurcation-analysis methods for machining dynamics.
- **Scientific programming and high-performance computing:** Advanced working knowledge of Python, including PyTorch and TensorFlow, MATLAB, and Mathematica. I use GPU-accelerated computing and parallel processing for neural-network training, numerical simulations, parameter studies, and large-scale analysis of nonlinear systems.
- **Experimental and data-analysis methods:** Practical experience with dynamometers, high-speed cameras, laser Doppler vibrometers, accelerometers, mechanical shakers, and data-acquisition systems. I use signal processing and experimental design to obtain, analyze, and interpret data for validation of computational models.
- **Research communication:** Peer-reviewed publications in journals including *Chaos*, *Chaos, Solitons & Fractals*, *The European Physical Journal Plus*, *The European Physical Journal B*, *Communications in Nonlinear Science and Numerical Simulation*, and *IFAC-PapersOnLine*. I have also presented research at international conferences and specialist meetings in nonlinear dynamics, control, and manufacturing systems.
- **Collaboration, teaching, and mentoring:** Experience collaborating across mathematics, engineering, AI, and experimental research. At UCD, I have co-mentored three Master's thesis projects involving turning-process dynamics and experimental modal analysis of medical handheld drills, and supported teaching in Applied Dynamics, Solid Mechanics, and Vehicle Structural Analysis.

Context & Impact

- My research sits at the intersection of nonlinear dynamics, scientific machine learning, numerical modeling, and applied mechanical systems. Across my publications, I have investigated how **data-driven** and **physics-informed** approaches can support the **prediction, reconstruction**, and analysis of complex behavior in nonlinear systems, including extreme events, chaotic attractors, soliton dynamics, and machining vibrations.
- My early research focused on predicting rare and potentially high-impact dynamical behavior. I developed machine-learning-based approaches for identifying the emergence of **extreme events** directly from system data. These studies demonstrated the potential of data-driven methods when governing equations are difficult to use in real time. I extended this work to chaotic dynamics by predicting **chaotic attractors** in the quasiperiodically forced logistic map using deep learning. This research explored whether deep learning can recover and predict complex attractor structures in a nonlinear discrete-time system. It strengthened my interest in computational approaches that infer dynamical features from data and in carefully validating learned models against numerical simulations.
- My work on **nonlinear Schrödinger-type partial differential equations** strengthened my connection between machine learning and differential-equation-based modeling. In my work on data-driven soliton solutions of the nonlinear Schrödinger equation with PT-symmetric potentials, I used deep learning to construct **soliton solutions** in nonlinear wave equations with parity-time-symmetric potentials. Subsequently, in my next work, I applied physics-informed neural networks to the Multi-Component Manakov Model, a higher-order coupled nonlinear partial differential equation model. This research showed how physical constraints and governing equations can be incorporated directly into neural-network training to reconstruct solutions from limited information.
- In my doctoral research, I extend physics-informed machine learning to parameter estimation in machining dynamics governed by regenerative delay and nonsmooth friction. In my manuscript Physics-Informed Neural Networks for Parameter Estimation in Machining Dynamics with Regenerative Delay and Nonsmooth Friction, I develop an **extended PINN** framework that infers unknown physical and process parameters from dynamical data while incorporating the machining model into learning. This inverse-problem setting seeks to recover parameters rather than simulate forward dynamics, despite challenges such as time-delay effects, discontinuous friction laws, nonlinear feedback, and **noisy measurements**. By combining experimental or simulated observations with physics-based residual constraints, the method aims to obtain parameter estimates consistent with both measured responses and the underlying mechanical model. This project builds directly on my earlier work with PINNs for nonlinear PDEs and data-driven dynamical modeling, forming a clear bridge between nonlinear dynamics, numerical modeling, machine learning, and inverse problems. Its impact includes improving identification of machining-system properties, understanding chatter, and enabling condition-aware manufacturing.
- My research has also been shaped by interdisciplinary collaboration. I have worked with researchers in nonlinear dynamics, computational mathematics, machine learning, manufacturing engineering, and experimental mechanics. These collaborations have enabled me to connect theoretical models with numerical algorithms, experimental data, and practical engineering questions.

Personal Reflection

- My research trajectory evolved from theoretical physics and nonlinear dynamical systems toward applied modeling of real engineering processes. Early work on **classical and quantum superintegrable systems** built my foundation in mathematical structures, differential equations, and **stability analysis** supporting my later transition into data-driven nonlinear dynamics and physics-informed machine learning.
- Working on extreme-event prediction and chaotic dynamics highlighted the limitations of purely analytical or purely data-driven approaches and motivated my interest in methods that combine observational data with mathematical structure and then I moved from idealized models to machining systems involving friction, time delay, stochasticity, nonsmooth interactions, and **experimental uncertainty**. Engaging with high-speed imaging, dynamometers, LDV, accelerometers, and **data-acquisition systems** strengthened my ability to connect models with physical measurements and to validate computational predictions against observed behavior.
- These projects required addressing challenges such as modeling delay and discontinuity, training neural networks for nonlinear differential equations, managing computational cost, and interpreting **noisy experimental data**. They reinforced my interest in inverse problems and in developing robust numerical and machine-learning methods by employing adaptive sampling and domain-based optimization for extracting meaningful dynamical information from incomplete or uncertain measurements.
- Presenting at **international conferences** helped refine how I communicate mathematical ideas to interdisciplinary audiences. Receiving the People's Choice Award in the UCD Thesis-in-3 competition strengthened my confidence in public communication. Peer-reviewing in AIP-Chaos has given me experience evaluating scientific research. Contributing to research-proposal writing has strengthened my ability to articulate motivation, feasibility, and impact in competitive funding contexts.

Conference Talks

- Dec 2023 Euromech Colloquium on Non-Smooth Dynamical Systems, University College Dublin, Ireland.
"Dynamics of Metal Cutting with Frictional, Regenerative, and Stochastic Effects"
- Jul 2024 11th European Nonlinear Oscillations Conference (ENOC), TU Delft, Netherlands.
"Study of Frictional and Regenerative Chattering in Metal Cutting with Stochastic Effects"
- Jun 2025 NODYCON 2025, Stevens Institute of Technology, Hoboken, NJ, USA.
"Data-driven Chatter Vibrations and Parameter Discovery in the Frictional and Regenerative Metal Cutting using Physics Informed Neural Networks" (Online)
- Aug 2025 17th IFAC Symposium on Large Scale Complex Systems: Theory and Applications, University College Dublin, Ireland.
"Experimental Insights into Orthogonal Metal Cutting via High-Speed Camera tests"
- Dec 2025 Euromech Colloquium on Recent Advances in Non-Smooth Dynamics, University of Exeter, United Kingdom.
"Experimental Characterization and Modelling of Medical Handheld Drills"
- Jul 2026 Conference on Nonlinear Systems and Dynamics (CNSD2026), IISER Trivandrum, India
"Multistability in Orthogonal Machining with Nonsmooth Frictional and Time-delay Effects"

Poster Presentations

- Jun 2023 SIAM UKIE Student Chapter, University of Oxford, United Kingdom.
"Discontinuity Induced Bifurcations in Machining Applications"
- Jun 2024 UCD STEM Symposium, Ireland.
"Basin Stability Analysis of Metal Cutting Dynamics"
- Jun 2025 UCD STEM Symposium, Ireland.
"Multi-stability in Orthogonal Machining with Regenerative, Frictional, and Stochastic Effects"
- Dec 2025 28th Sir Bernard Crossland Symposium, University College Dublin, Ireland.
"Multi-stability in Orthogonal Machining with Regenerative, Frictional, and Stochastic Effects"

Awards & Grants

1. SIAM UKIE Travel Grant – To attend the SIAM UKIE student chapter at the University of Oxford, United Kingdom. (2023)
2. UCD CEA Seed Funding: Dissemination & Engagement Award (2026)
3. UCD Advantage Award (2026)
4. People's Choice Award – UCD ThesisIn3 Competition (2026)

Volunteering Activities

1. Leading Volunteer during the 72nd General Assembly of the International Academy for Production Engineering (CIRP) that took place in the University College Dublin campus from August 20th to August 26th, 2023.
2. Member of the Local Organizing Committee for the Euromech Colloquium on Non-smooth Dynamical Systems in UCD from 11th December 2023 to 13th December 2023.
3. Volunteer for UCD STEM Early Career Researchers Symposium 2024.
4. Volunteer for IFAC Symposium on Large Scale Complex Systems (Aug 2025).
5. PhD Student Representative - Equality, Diversity, and Inclusion (EDI), School of Mechanical and Materials Engineering, UCD.

Referees

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